Supplementary Material S1

Definitions, Sources, Selection Criteria, Scoring Rules and Additional Results for the Composite Vulnerability Index

This supplementary material accompanies “Composite Vulnerability Index of Villages with Oil Palm on Peatland in Indonesia in 2021: Integrating Remote Sensing and Village Development Index Data.” It documents every variable used to construct the Exposure, Sensitivity, and Adaptive Capacity pillars of the Composite Vulnerability Index (CVI), maps each dataset field to its official 2021 Village Development Index (Indeks Desa Membangun, IDM) indicator number and scoring rule, lists the IDM indicators that were excluded with the rationale for each exclusion, and reports the additional robustness and bias checks cited in the main text (Tables S5–S9, Figure S1). Indicators were retained only if they met three criteria: (i) a conceptual link to sensitivity or adaptive capacity in the Exposure-Sensitivity-Adaptive Capacity framework (Cutter et al., 2003) and in oil palm livelihood studies; (ii) an unambiguous direction, with a higher raw score indicating a more favorable condition; and (iii) availability for the analytic sample. Of the 50 IDM items, 34 were retained (8 in Sensitivity, Table S2; 26 in Adaptive Capacity, entered as 24 fields, Table S3) and 16 were excluded (Table S4); 34 + 16 = 50. All IDM indicators are scored on a 0-5 scale following the Kemendes PDTT Standard Operating Procedure; scoring summaries below are abbreviated and should be read alongside the full SOP for exact thresholds.

Files supplied with this article

All supplementary materials are hosted in the public repository https://github.com/willyonesian29/cvi-peatland-oilpalm-2021.

File	Content
Supplementary Material S1 (this document; supplementary/ folder)	Detailed methods, variable definitions, selection rules and additional tables (S1–S9) and Figure S1
Supplementary Material S2 (data/S2_village_cvi_scores.csv)	Village-level records for all 901 villages: province, regency, district, village code and name, Ina-Geoportal village area, oil palm area on peat (total, industrial, smallholder), exposure ratio, normalized Exposure, Sensitivity and Adaptive Capacity, CVI, Jenks class, dominant plantation type, priority-group membership, consensus flag, Monte Carlo top-group frequency, priority type and LISA cluster class
Supplementary Material S3 (code/ folder)	The Earth Engine script for oil palm on peat (00b), the village overlay (00b2), Python scripts for the analysis input (00d), consistency checks (00c), baseline index (01), robustness, fixed-effects and spatial analyses (02), priority-group stability and bias checks (03), Tables S5–S6 (04) and Figures 1, 2 and S1 (05–07); also at the public repository https://github.com/willyonesian29/cvi-peatland-oilpalm-2021

Detailed Methods

The main text gives a condensed description of the Materials and Methods, as required by the journal’s word limit. This section gives the full description.

Study Area and Data

The study was conducted from July to August 2026 at the Urban and Regional Planning Master Programme, Diponegoro University, and the Graduate School of Innovation and Practice for Smart Society, Hiroshima University, using secondary spatial and socioeconomic data for the 2021 reference year covering Indonesian villages (desa/kelurahan) whose administrative boundaries contain oil palm on peatland. Oil palm extent and plantation type (industrial or smallholder) were taken from the 2021 layer of the global oil

palm dataset of Descals et al. (2024), mapped at 10 m resolution from Sentinel-1 imagery, and peatland from the national thematic peatland map (scale 1:50,000) distributed through One Map Indonesia by the Geospatial Information Agency (BIG). Village boundaries were the Ina-Geoportal village layer of BIG (1:10,000 base-map series, edition of 28 September 2023). In Google Earth Engine, the peatland polygons were rasterized to a 10 m mask, oil palm pixels on peat were selected separately for industrial and smallholder plantations and converted to polygons at the native 10 m resolution. These were overlaid with the village polygons in a desktop GIS, and their areas, calculated on the WGS 84 ellipsoid, were summed per village; village area was computed from the same polygons, so that the numerator and denominator of Exposure share one boundary set. The results were joined to the IDM records by Kemendagri village code (KDEPUM; the IDM code is the same number without separators). Each layer carries uncertainty. For 2021, the producer's and user's accuracies of the oil palm map are about 91% for industrial plantations but only 71-72% for smallholdings (Descals et al., 2024), so smallholdings are partly omitted or confused with industrial blocks; peat boundaries at 1:50,000 are generalized and the map used here does not separate depth classes; and the 2023 boundary edition, mostly delineated in 2018-2019, may differ locally from the boundaries that village governments reported for 2021. The analytic sample was restricted to villages with non-zero oil palm area on peatland (n = 923); 22 villages with duplicate administrative codes were removed, leaving 901 villages across 11 provinces. The workflow is summarized in Figure 1 of the main text.

Socioeconomic indicators were obtained from the raw, indicator-level 2021 IDM dataset of the Ministry of Villages, Disadvantaged Regions Development, and Transmigration, comprising 50 indicators scored 0-5 across Social (35), Economic (12) and Environmental (3) Resilience (Kemendes PDTT, 2021). The data are reported by village governments and verified by district facilitators, so they describe the availability of services rather than household outcomes. Indicators were selected with three explicit criteria: (i) a conceptual link to sensitivity or adaptive capacity in the vulnerability framework (Cutter et al., 2003) and in oil palm livelihood studies (Krishna et al., 2017; Santika et al., 2019); (ii) an unambiguous direction, with higher raw scores indicating better conditions; and (iii) availability for the sample. Sensitivity indicators describe conditions that determine how strongly households are harmed when drainage, flooding, fire or pollution occur: reliance on unprotected water sources and poor sanitation, existing pollution, disaster incidence and social conflict, and low economic diversity, i.e., dependence on few land-based incomes. Adaptive Capacity indicators describe the resources that allow a village to absorb or adjust to these pressures: health and education services, electricity and connectivity, markets, finance, transport and local institutions. Thirty-four of the 50 items were retained: 8 as Sensitivity and 26 as Adaptive Capacity items, the latter entered as 24 fields because three health-worker items are combined. Two binary items from the same questionnaire that are not part of the IDM score, the presence of a village-owned enterprise (BUMDes) and of a farmer group, were added to Adaptive Capacity as institutional-capacity measures. Sixteen items (e.g., religious, linguistic and ethnic diversity, sports groups, security posts) were excluded for lack of a plausible link. Table 1 of the main text summarizes the structure, and Tables S1–S4 list every variable with its IDM number, scoring rule, direction and exclusion rationale. Content validity therefore rests on these documented criteria; because no outcome data on peatland-related harm were available, the index could not be validated externally (Rufat et al., 2019). None of the 32 IDM fields had missing values, and no indicator pair within a pillar exceeded $|r| = 0.80$.

Index Construction

Exposure (E) was the ratio of oil-palm-on-peatland area to village area. It measures how much of the village territory is under drainage-based cultivation on peat and therefore subject to subsidence, fire and flood hazards that extend beyond plantation boundaries (Hooijer et al., 2012; Sumarga et al., 2016). It is an extent measure: it does not capture peat depth, drainage intensity or degradation, nor how far household incomes depend on oil palm, because consistent village-level data on these were not available for the sample. Three villages, Sumber Makmur and Makarti Jaya in Nagan Raya (Aceh) and Pawalutan in Hulu Sungai Utara (Kalimantan Selatan), had ratios above 1 (1.56-2.18), a geometry artefact of the overlay, and were capped at 1.0; their influence was tested in the robustness analysis (Table 5 of the main text). Sensitivity (S) and Adaptive Capacity (AC) were unweighted means of their indicators, with Sensitivity reversed so that higher values denote greater sensitivity. Each pillar was min-max normalized to 0-1, and the CVI was computed as:

$$CVI = [E + S + (1 - AC)] / 3$$

Equal weighting was the default in the absence of an empirical or theoretical basis for differential weights (OECD et al., 2008). Villages were classified into five classes (Very Low to Very High) with Jenks natural breaks (k = 5, fixed random seed); the upper class bounds were 0.186 (Very Low), 0.277 (Low), 0.363 (Moderate), 0.467 (High) and 0.713 (Very High). Jenks breaks are specific to this sample, so the classes describe a village's relative position among the 901 villages rather than an absolute level of vulnerability.

Robustness and Statistical Analysis

Because nominal weights do not guarantee equal influence of each pillar (Becker et al., 2017; Greco et al., 2019), the CVI was recomputed under nine alternatives: principal component analysis (PCA) weights; exposure-weighted (0.50/0.25/0.25) and socioeconomic-weighted (0.20/0.40/0.40) schemes; geometric aggregation, which limits compensation between pillars; rank-normalized exposure, which removes the effect of its right skew; exposure winsorized at the 99th percentile; exclusion of the three capped villages; removal of the two non-IDM binaries; and leave-one-indicator-out (34 runs, one for each Sensitivity and Adaptive Capacity field including the two binaries). In addition, 1,000 weight vectors were drawn from a flat Dirichlet distribution (Monte Carlo). Each variant was compared with the baseline by Spearman's rho, the share of the baseline High/Very High villages (the priority group, n = 207) remaining among the 207 highest-ranked villages, the median absolute change in rank, and whether the smallholder-industrial difference kept its sign and significance. Because rank correlations can remain high while the villages at the top change, the priority group was also evaluated directly: (i) agreement between baseline classes and Jenks classes recomputed for each variant (quadratic-weighted Cohen's kappa and share of villages changing class); (ii) a consensus priority set, defined as priority villages that remain among the 207 highest-ranked under all eight single alternatives and in at least 90% of Monte Carlo draws; and (iii) the pillar profile of villages entering and leaving the priority group. Priority villages were further split into exposure-led (exposure ratio at or above the sample 75th percentile) and development-deficit (below the median) types. The effective influence of each pillar was summarized by its squared rank correlation with the CVI (Becker et al., 2017).

Provincial differences were tested with the Kruskal-Wallis test (provinces with n ≥ 10), and smallholder- versus industrial-dominated villages (defined by the larger plantation-type area of oil palm on peat) with the Mann-Whitney U test, repeated for each pillar and in an ordinary least squares model with province fixed effects and standard errors clustered by regency. Because villages are nested within regencies, the share of CVI variance lying between regencies was also estimated as the intraclass correlation coefficient (ICC) of a random-intercept model. Spatial autocorrelation of the CVI, Exposure and the fixed-effects residuals was tested with global Moran's I and local indicators of spatial association (LISA) (Anselin, 1995), using an eight-nearest-neighbour, row-standardized weight matrix of the centroids of the 901 Ina-Geoportal village polygons and 999 permutations. Because the fixed-effects residuals were spatially autocorrelated, the industrial-dominance association was re-estimated with a maximum-likelihood spatial error model with province fixed effects (Anselin, 1988). To assess bias from the plantation-type classification and from regional differences, the comparison was repeated excluding mixed villages (smallholder share of oil palm on peat 40-60%), with a two-thirds dominance threshold, and with log village area and log oil palm area on peat as additional controls in the fixed-effects model. Analyses used Python 3 (pandas, scikit-learn, scipy, statsmodels, mapclassify, geopandas, libpysal, esda, spreg). Village-level pillar scores, CVI values and classes (Supplementary Material S2) and the complete Python and Google Earth Engine code (Supplementary Material S3) are provided with the article and in a public repository (https://github.com/willyonesian29/cvi-peatland-oilpalm-2021) so that all results can be reproduced.

Table S1. Exposure Variable: Definition, Spatial Derivation and Uncertainty

Dataset Field	Definition	Source / Derivation	Treatment in CVI
peat_tp_ha	Oil palm area located on peatland within the village (hectares)	Google Earth Engine: the BIG peatland map (1:50,000) rasterized to a 10 m mask; oil palm pixels of the 2021 layer of Descals et al. (2024) on peat selected and converted to polygons at 10 m (00b). These were overlaid with the Ina-Geoportal	Numerator of exposure_ratio

Dataset Field	Definition	Source / Derivation	Treatment in CVI
		village polygons in a desktop GIS and their ellipsoidal (WGS 84) areas summed per village (00b2). peat_tp_ha = peat_ip_ha + peat_sp_ha	
VillageArea_Ha	Village area (hectares)	Geodesic area (WGS 84) of the same Ina-Geoportal village polygon (BIG village boundaries, 1:10,000, edition 28 September 2023); identical to the layer's LUASWH field	Denominator of exposure_ratio
exposure_ratio	peat_tp_ha ÷ VillageArea_Ha, capped at 1.0	Derived	Min-max normalized to produce Exposure (0-1)
peat_ip_ha	Industrial oil palm area on peatland (hectares)	Same overlay, stratified by plantation type per Descals et al. (2024)	Defines dominant_type (industrial- vs smallholder-dominated: larger of peat_ip_ha and peat_sp_ha); not part of the Exposure score
peat_sp_ha	Smallholder oil palm area on peatland (hectares)	Same overlay, stratified by plantation type	Defines dominant_type and the smallholder share (peat_sp_ha ÷ peat_tp_ha)

Note on uncertainty: for the 2021 layer, Descals et al. (2024) report producer's and user's accuracies of 91.0 ± 2.5% and 91.8 ± 1.2% for industrial plantations and 71.4 ± 0.7% and 72.4 ± 1.8% for smallholder plantations. Plantation type is assigned from image morphology (size, shape and road pattern), not legal tenure, so plasma and organized smallholdings may be recorded as industrial. The 1:50,000 peat map does not separate depth classes and its boundaries are generalized. The village boundaries are the 2023 edition of the Ina-Geoportal layer, delineated mainly in 2018–2019 (865 of the 901 villages), and may differ locally from the boundaries that villages reported for 2021. Numerator and denominator of the exposure ratio come from the same polygons. In three villages (Sumber Makmur and Makarti Jaya, Nagan Raya; Pawalutan, Hulu Sungai Utara) oil palm on peat exceeded the polygon area (ratios 1.56–2.18), a geometry artefact of the overlay; these were capped at 1 and excluded in a sensitivity test (Table 3 of the main text).

Table S2. Sensitivity Indicators (n = 8)

All eight indicators were reverse-coded (Sensitivity contribution = 1 − normalized raw score) so that a higher value indicates greater sensitivity, consistent with their original scoring direction in which a higher raw IDM score denotes a more favorable condition.

Dataset Field	Official IDM Indicator (No.)	Original 0-5 Scoring Rule (summary)	Dimension
air_minum	Sumber Air Minum (#34)	PAM/piped water=5; borewell/pump=4; bottled/spring=3; river/rain=2; other=1	IKS
air_mck	Air Mandi & Cuci (#35)	PAM/piped water=5; borewell/pump=4; protected spring=3; river/rain=2; other=1	IKS
jamban	Akses Jamban (#32)	Private toilet=5; shared=4; public=3; other=0	IKS
tps	Sampah / Waste Management (#33)	Collected by service=5; buried/burned=4; river/drain=1; other=0	IKS
konflik	Konflik (#25)	No mass conflict=5; conflict present=1	IKS

Dataset Field	Official IDM Indicator (No.)	Original 0-5 Scoring Rule (summary)	Dimension
cemar	Kualitas Lingkungan (#48)	No pollution=5; 0.25 severity=4; 0.5=3; 0.75=2; severe/masif=0	IKL
bencana	Rawan Bencana (#49)	0 disaster types=5; 1=4; 2=3; ≥3 types (incl. karhutla/fire)=0	IKL
ragam_eko	Keragaman Produksi (#36)	Micro-industry-to-household ratio ≥0.004=5; 0.001-0.003=3; 0=1	IKE

Table S3. Adaptive Capacity Indicators (n = 26)

The first 24 fields are official IDM indicators used directly (higher raw score = greater capacity, no inversion applied); because the field nakes combines three health-worker items (#2-4), these 24 fields cover 26 IDM items. The final two fields (bumdes_bin, l_tani_bin) are descriptive variables collected in the same Kemendes PDTT questionnaire but excluded from the official IDM score formula; they were added because they measure local institutional capacity, a core construct of the Exposure-Sensitivity-Adaptive Capacity framework (Cutter et al., 2003). Their influence was tested by removing both (Table 3 of the main text).

Dataset Field	Official IDM Indicator (No.)	Original 0-5 Scoring Rule (summary)	Dimension
faskes_km	Akses Sarana Kesehatan (#1)	Travel time to nearest health facility: ≤30min=5; 31-60=4; 61-90=3; 91-120=2; >120=1	IKS
nakes	Tenaga Kesehatan (#2-4, combined proxy)	Combined availability of doctors, midwives, and other health workers	IKS
posyandu	Akses Poskesdes (#6)	Distance to nearest community health post: ≤500m=5 … >3500m=1	IKS
posyandu_aktif	Aktivitas Posyandu (#7)	Ratio of posyandu active ≥1x/month: >0.75=5 … 0=1	IKS
bpjs	Kepesertaan BPJS (#5)	BPJS health-insurance participation ratio: >0.75=5 … 0=1	IKS
sd	Akses SD/MI (#8)	Distance to nearest primary school: ≤3000m=5 … >10000m=1	IKS
smp	Akses SMP/MTs (#9)	Distance to nearest junior secondary school: ≤6000m=5 … ≥12000m=1	IKS
sma	Akses SMU/SMK (#10)	Distance to nearest senior secondary school: ≤6000m=5 … ≥12000m=1	IKS
kursus	Akses Pusat Keterampilan (#13)	Availability of vocational/skills training center: present=5; absent=1	IKS
listrik_kk	Akses Listrik (#28)	Household electrification ratio: ≥0.90=5 … <0.50=1	IKS
sinyal	Sinyal Telepon Seluler (#29)	Cellular signal strength: strong=5; weak=3; none=0	IKS
inet_desa	Internet Kantor Desa (#30)	Internet availability at village office: present=5; absent=1	IKS
inet_warga	Akses Internet Warga (#31)	Internet access available to residents: present=5; absent=1	IKS
mitigasi	Tanggap Bencana (#50)	No. of mitigation facilities (early warning, evacuation route, safety equipment): ≥3=5 … 0=0	IKL

Dataset Field	Official IDM Indicator (No.)	Original 0-5 Scoring Rule (summary)	Dimension
pusat_dagang	Akses Pertokoan (#37)	Distance to nearest shopping cluster: ≤7km=5 … >25km=1	IKE
pasar	Akses Pasar (#38)	Households-per-market ratio: ≥250=5; <250=3; none=1	IKE
minimarket	Toko & Warung Kelontong (#39)	No. of retail/grocery shops: >3=5; 3=4; 2=3; 1=2; 0=1	IKE
logistik	Pos & Jasa Logistik (#41)	No. of postal/courier service points: >1=5; 1=3; 0=0	IKE
bank	Bank & BPR (#42)	No. of bank/rural credit branches: >1=5; 1=3; 0=1	IKE
kredit	Fasilitas Kredit (#43)	No. of accessible credit-scheme types: 4=5; 3=4; 2=3; 1=2; 0=1	IKE
koperasi	Lembaga Ekonomi (#44)	No. of active cooperatives/BUMDes: >1=5; 1=3; 0=1	IKE
angkot	Moda Transportasi Umum (#45)	Public transport: fixed route=5; no fixed route=3; none=1	IKE
jalan_mobil	Keterbukaan Wilayah (#46)	4-wheel vehicle road access: year-round=5; seasonal=3; impassable=1	IKE
jalan_mutu	Kualitas Jalan (#47)	Road surface: asphalt/concrete=5; gravel=4; dirt=3; other=1	IKE
bumdes_bin	Not part of official IDM score — Questionnaire item 425f	Presence of village-owned enterprise (BUMDes): recoded Ada=1, Tidak Ada=0	Supplementary (binary)
l_tani_bin	Not part of official IDM score — Questionnaire item 341f	Presence of a farmer group: recoded Ada=1, Tidak Ada=0	Supplementary (binary)

Table S4. Official IDM Indicators Considered but Excluded from the CVI (11 groups covering 16 of the 50 original items)

Indicators were excluded where theoretical review found no plausible conceptual link to peatland-oil-palm exposure, environmental/economic sensitivity, or adaptive capacity, following the selection principle described in the Materials and Methods section.

Official IDM Indicator (No.)	Rationale for Exclusion
Ketersediaan PAUD (#11)	Early-childhood education access; no plausible conceptual link to peatland-oil-palm exposure, sensitivity, or adaptive capacity
Ketersediaan PKBM/Paket ABC (#12)	Non-formal education package program; conceptually distal to the study's environmental-economic vulnerability framework
Taman Bacaan/Perpustakaan (#14)	Library/reading-room access; no direct theoretical link to the framework
Kebiasaan & Frekuensi Gotong Royong (#15, #16)	General community mutual-aid custom; not specific to environmental or economic adaptive capacity in this context
Keberadaan Ruang Publik (#17)	Public open-space availability; no direct conceptual link
Kelompok & Kegiatan Olahraga (#18, #19)	Sports group/activity presence; no plausible link to peatland-palm-oil vulnerability

Official IDM Indicator (No.)	Rationale for Exclusion
Keragaman Agama/Bahasa/Suku (#20, #21, #22)	Religious, linguistic, and ethnic diversity measure general social heterogeneity, not vulnerability to land-use-driven environmental degradation
Pos Keamanan & Siskamling (#23, #24)	General community security infrastructure; not conceptually tied to environmental/economic sensitivity or capacity
PMKS (#26)	Count of social-welfare problem cases; indirect and non-specific link to the exposure-sensitivity-capacity framework; excluded for parsimony
Akses SLB (#27)	Special-needs school access; no direct conceptual link
Kedai & Penginapan (#40)	Food-stall/lodging availability functions primarily as a tourism proxy rather than a livelihood-diversification or adaptive-capacity measure

Notes

1. Indicator numbering follows the 2021 Panduan Variabel dan Aturan Skoring IDM (Kemendes PDTT). Where a dataset field combines more than one official indicator (e.g., nakes), this is noted explicitly rather than implying an exact one-to-one correspondence.

2. None of the 32 IDM fields had missing values in the analytic sample of 901 villages; the median-imputation step in the code is retained only as a safeguard and changes no value.

3. All pillar-level scores (Exposure, Sensitivity, Adaptive Capacity) were min-max normalized to a 0-1 range after aggregation. The CVI was computed as the equally weighted average of Exposure, Sensitivity, and (1 − Adaptive Capacity). Jenks natural breaks (k = 5, random seed 42) were applied to the equally weighted CVI; class bounds are given in Table 5 of the main text.

Table S5. Leave-One-Indicator-Out Results

Rows are sorted from the most to the least influential field. rho = Spearman correlation with the baseline CVI; Priority group retained = share of the 207 baseline High/Very High villages still among the 207 highest-ranked villages.

Removed field	Pillar	Spearman rho	Priority group retained (%)	Median absolute rank change	Smallholder < industrial	Mann-Whitney p
ragam_eko	Sensitivity	0.958	83.6	52	Yes	6.5e-18
tps	Sensitivity	0.982	90.8	20	Yes	7.4e-16
konflik	Sensitivity	0.987	96.6	13	Yes	5.0e-15
cemar	Sensitivity	0.990	93.7	20	Yes	4.4e-14
air_minum	Sensitivity	0.991	95.7	17	Yes	2.9e-16
air_mck	Sensitivity	0.993	94.7	15	Yes	2.8e-15
bencana	Sensitivity	0.993	95.2	17	Yes	2.1e-15
jamban	Sensitivity	0.994	95.7	6	Yes	2.0e-15
sma	Adaptive Capacity	0.997	96.1	13	Yes	3.6e-15
mitigasi	Adaptive Capacity	0.997	96.6	17	Yes	5.9e-16
logistik	Adaptive Capacity	0.997	96.6	12	Yes	1.9e-16

Removed field	Pillar	Spearman rho	Priority group retained (%)	Median absolute rank change	Smallholder < industrial	Mann-Whitney p
pasar	Adaptive Capacity	0.997	97.6	13	Yes	8.8e-16
inet_warga	Adaptive Capacity	0.997	96.6	13	Yes	6.1e-16
inet_desa	Adaptive Capacity	0.997	96.6	14	Yes	1.2e-15
pusat_dagang	Adaptive Capacity	0.997	96.1	16	Yes	5.4e-16
bumdes_bin	Adaptive Capacity	0.997	97.1	7	Yes	2.2e-15
jalan_mobil	Adaptive Capacity	0.998	95.7	9	Yes	9.3e-16
koperasi	Adaptive Capacity	0.998	97.6	8	Yes	1.0e-15
angkot	Adaptive Capacity	0.998	97.1	12	Yes	6.2e-16
kursus	Adaptive Capacity	0.998	99.0	7	Yes	2.7e-15
bank	Adaptive Capacity	0.998	98.1	6	Yes	1.8e-15
posyandu	Adaptive Capacity	0.998	97.6	11	Yes	5.0e-16
l_tani_bin	Adaptive Capacity	0.998	97.6	5	Yes	3.9e-16
bpjs	Adaptive Capacity	0.999	98.1	8	Yes	2.2e-15
listrik_kk	Adaptive Capacity	0.999	97.1	4	Yes	2.0e-15
minimarket	Adaptive Capacity	0.999	98.1	4	Yes	7.6e-16
kredit	Adaptive Capacity	0.999	98.6	7	Yes	6.6e-16
sinyal	Adaptive Capacity	0.999	97.1	8	Yes	1.9e-15
jalan_mutu	Adaptive Capacity	0.999	98.6	5	Yes	1.9e-15
smp	Adaptive Capacity	0.999	98.1	4	Yes	9.1e-16
nakes	Adaptive Capacity	0.999	98.1	5	Yes	1.2e-15
sd	Adaptive Capacity	1.000	100.0	0	Yes	1.1e-15
posyandu_aktif	Adaptive Capacity	1.000	100.0	1	Yes	1.3e-15

Removed field	Pillar	Spearman rho	Priority group retained (%)	Median absolute rank change	Smallholder < industrial	Mann-Whitney p
faskes_km	Adaptive Capacity	1.000	100.0	0	Yes	1.2e-15

Table S6. Pillar Decomposition of the Smallholder-Industrial Difference

Normalized pillar scores (0-1). Dominant type is defined by the larger plantation-type area of oil palm on peat in the village.

Pillar	Median, smallholder-dominated (n = 421)	Median, industrial-dominated (n = 480)	Mann-Whitney U	p
Exposure (normalized)	0.003	0.081	35,924	1.2e-62
Sensitivity	0.227	0.273	90,832	8.7e-03
1 − Adaptive Capacity	0.487	0.513	93,974	7.0e-02
CVI	0.256	0.312	69,861	1.2e-15

Table S7. LISA Cluster Counts by Province, Regency and Priority Status

Local Moran's I of the CVI for all 901 villages, using the centroids of the Ina-Geoportal village polygons, eight-nearest-neighbour row-standardized weights and 999 permutations; $p < 0.05$. High-High villages had a median CVI of 0.399 and Low-Low villages 0.182.

Panel A. Cluster type by province

Province	High-High	Low-Low	High-Low	Low-High	Not significant	Total
Kalimantan Tengah	74	26	4	14	119	237
Riau	29	32	6	4	88	159
Sumatera Selatan	14	0	0	0	22	36
Kalimantan Barat	12	71	15	5	207	310
Aceh	10	0	0	1	35	46
Sumatera Utara	6	0	0	3	5	14
Kalimantan Utara	2	0	0	1	17	20
Jambi	0	7	1	0	0	8
Kalimantan Selatan	0	2	1	0	36	39
Kalimantan Timur	0	0	0	0	16	16
Sumatera Barat	0	0	0	1	15	16
Total	147	138	27	29	560	901

Panel B. Regencies with the most High-High villages

Regency	Province	Villages	Median CVI
Kapuas	Kalimantan Tengah	41	0.430
Indragiri Hulu	Riau	15	0.405

Regency	Province	Villages	Median CVI
Kotawaringin Timur	Kalimantan Tengah	15	0.411
Indragiri Hilir	Riau	12	0.386
Nagan Raya	Aceh	10	0.465
Katingan	Kalimantan Tengah	7	0.350
Barito Selatan	Kalimantan Tengah	4	0.394
Tapanuli Tengah	Sumatera Utara	4	0.522

Panel C. Regencies with the most Low-Low villages

Regency	Province	Villages	Median CVI
Bengkalis	Riau	30	0.183
Kotawaringin Barat	Kalimantan Tengah	23	0.155
Kubu Raya	Kalimantan Barat	18	0.113
Sekadau	Kalimantan Barat	14	0.209
Mempawah	Kalimantan Barat	12	0.146

Panel D. Cluster type by priority status (see Table 4 of the main text)

Priority status	High-High	Low-Low	High-Low	Low-High	Not significant	Total
Consensus priority	35	0	0	0	19	54
Other High/Very High	62	0	10	0	81	153
Not priority	50	138	17	29	460	694

Table S8. Checks for Plantation-Type Classification and Regional Bias in the Smallholder-Industrial Comparison

OLS models include province fixed effects; standard errors are clustered by regency. The dependent variable is the CVI unless a pillar is named; for Adaptive Capacity a negative β means lower capacity in industrial-dominated villages. Because oil palm area on peat is the numerator of Exposure, the model with area controls largely removes the exposure difference itself.

Check	n	β (industrial-dominated)	SE	p
Mixed villages (smallholder share of oil palm on peat 40-60%) excluded	858	0.066	0.012	2.3e-08
Two-thirds dominance threshold (Mann-Whitney; median CVI 0.255 vs 0.312)	832			8.5e-16
Controls: log village area, log oil palm area on peat	901	0.024	0.012	4.0e-02
Pillar S with same controls	901	0.012	0.016	4.3e-01
Pillar AC with same controls	901	-0.036	0.022	1.0e-01

Table S9. Class Transition Matrices between Baseline and Variant Jenks Classes

Jenks breaks (k = 5) were re-applied to each variant and compared with the baseline classes. Panel A summarizes agreement for all eight single alternatives; Panel B gives the full transition matrix for geometric

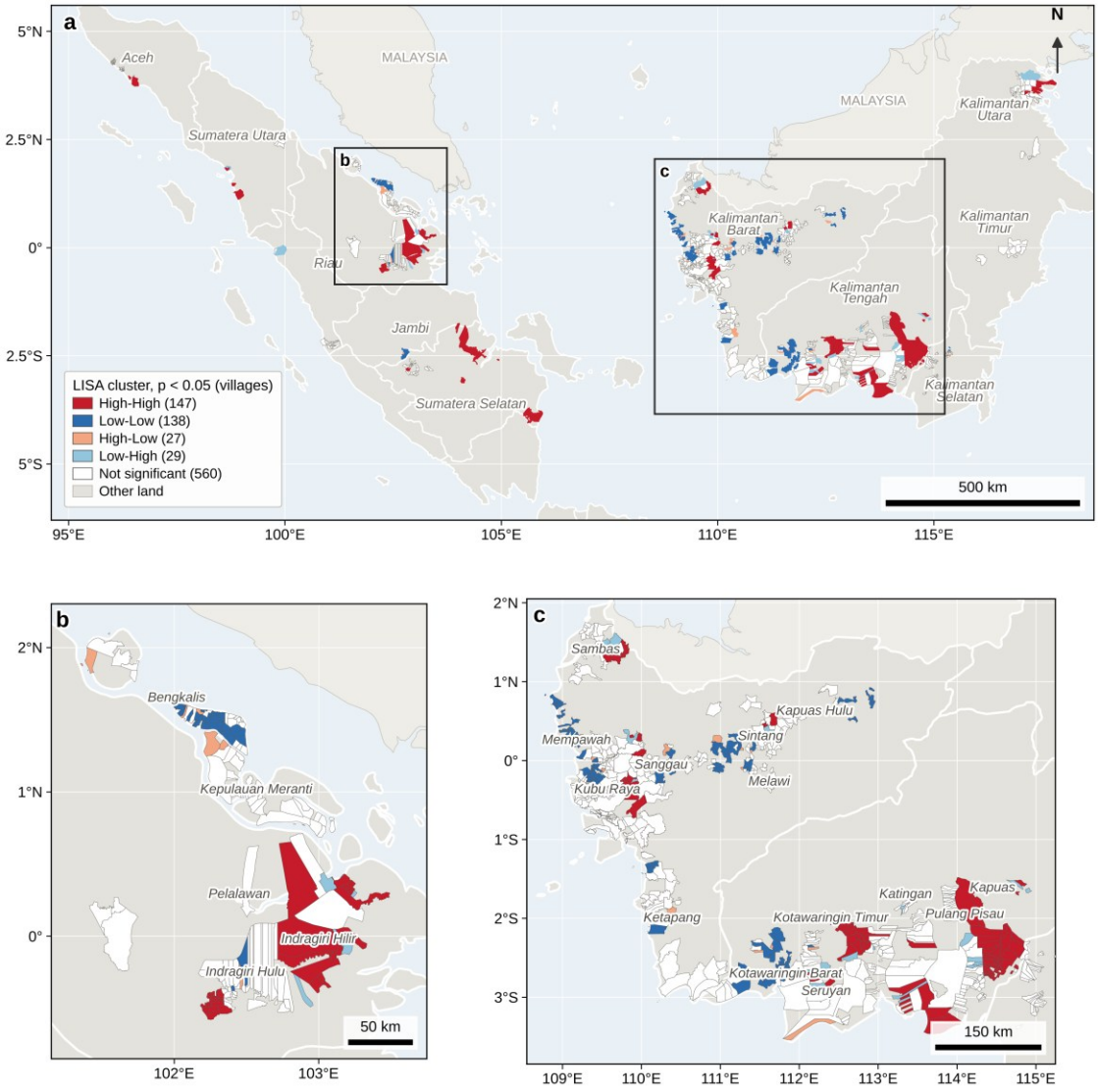
aggregation, the least concordant variant. Matrices for all variants are in tableS9_class_transitions.csv (Supplementary Material S3).

Panel A. Agreement with baseline classes

Variant	Quadratic-weighted kappa	Villages changing class (%)	Changing more than one class (%)
Geometric aggregation	0.691	51.1	9.2
PCA-derived weights	0.789	47.3	2.3
Rank-normalized Exposure	0.808	51.4	0.2
Socioeconomic-weighted (0.20/0.40/0.40)	0.871	34.5	0.1
Exposure-weighted (0.50/0.25/0.25)	0.882	29.6	0.0
Without BUMDes and farmer-group items	0.915	23.8	0.0
Exposure winsorized (99th pct)	0.957	11.0	0.0
Capped village excluded	0.986	3.8	0.0

Panel B. Baseline class (rows) × class under geometric aggregation (columns), number of villages

Baseline class	Very Low	Low	Moderate	High	Very High
Very Low	131	23	0	0	0
Low	104	117	57	0	0
Moderate	20	114	96	32	0
High	0	44	29	62	7
Very High	0	10	9	11	35



Remarks: Local Moran's I of the CVI with an eight-nearest-neighbour, row-standardized weight matrix of the centroids of the 901 Ina-Geoportal village polygons and 999 permutations; $p < 0.05$. Village polygons from the Ina-Geoportal village boundaries of BIG (2023 edition); province outlines and neighbouring countries from Natural Earth (1:10 million). Geographic coordinates, WGS 84.

Figure S1. Local Indicators of Spatial Association (LISA) Clusters of the CVI in 2021: (a) All Study Villages, (b) Riau, (c) Kalimantan Barat and Kalimantan Tengah

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